

DO NOT WRITE ON TEST



**COMPUTER
PROGRAMMING
CONCEPTS
~OPEN EVENT~**

(390)

STATE 2023

TOTAL POINTS (50 @ 2pts each) _____100 Points

Test Time: 60 minutes

Directions: Identify the letter of the choice that *best* completes the statement or answers the question.

1. The number system used by computers that uses digits 0 to 1 is known as what?
 - A. Base-2 #
 - B. Machine Code
 - C. Hexadecimal
 - D. Base-8
2. In the Windows-1252 default character subset, which entity numbers are the exact same as the original ASCII character sets?
 - A. 0 to 127
 - B. 32 to 127
 - C. 0 to 31
 - D. 32 to 64
3. How many Kilobyte in binary does it take to create Gigabyte?
 - A. 1,024 Megabytes
 - B. 1,024 Kilobytes
 - C. 1,048,576 Bytes
 - D. 1,048,576 Kilobytes
4. What was one of the motivations to create Unicode versus just using ASCII?
 - A. Unicode can be highly optimized for internet transmissions
 - B. ASCII does not properly store in RAM causing programming issues
 - C. ASCII only has 128 characters definitions and does not support international community #
 - D. ASCII is not cross platform
5. What best explains the functions of the ALU in a CPU?
 - A. Holds the data that has been retrieved out of memory
 - B. In charge of all instruction registrations
 - C. In charge of all math and logic calculations
 - D. In charge of all operations that need to be carried out and communicated with the memory
6. Which of the following terms best fits the description that it provides temporary storage for the CPU to access information stored on the hard drive?
 - A. RAM
 - B. ROM
 - C. CPU Register #
 - D. Arithmetic Operator

7. This type of software allows the user to create or alter raster or vector images:
 - A. Spreadsheet software
 - B. Presentation software
 - C. Graphic software #
 - D. Database software
8. This type of programming language is characterized by creating instances of other classes as elements of the program to store and control information, thus allowing it a modular approach and pooling multiple elements together to solve specific needs:
 - A. Procedural
 - B. Functional
 - C. Object Oriented
 - D. Scripting
9. Which of the following is a characteristic of a low-level programming language?
 - A. Programs in natural language
 - B. Allows user to make changes to code while program is running
 - C. Program instructions very close to machine instructions
 - D. Contains strong abstraction
10. Which type of network uses an implementation of a spoke-hub design where all hosts are connected to a central hub?
 - A. Star Network
 - B. Bus Network
 - C. Tree Network
 - D. Centralized Network
11. The data in a class could also be called which of the following?
 - A. Attributes
 - B. Fields
 - C. Instance Variables
 - D. All of the above
12. When a programmer strategically places data within methods that processes that data in a single unit?
 - A. Divide and conquer
 - B. Encapsulation
 - C. Inheritance
 - D. Polymorphism
13. Which definition best describes a method in Java?
 - A. One occurrence of a common data type
 - B. Class variables that store information
 - C. Behavior based components that process data
 - D. Breaking abstract details into less complicated forms

14. Which of the following is NOT a characteristic of a constructor?
- A. Is called when an object is being instantiated
 - B. It is not a void method
 - C. It does not return any data
 - D. Its name is unique and different than the class identifier
15. In Java, when is the constructor of a super class called?
- A. It is called automatically during the creation of a subclass object
 - B. It is called automatically during the creation of a super class object
 - C. It is called automatically during the creation of a static class object
 - D. It can only be called if the program uses the word 'super'
16. When methods of a class are declared private, which classes can access them?
- A. Members of any class anywhere
 - B. Only members of any class in the same package
 - C. Members of the same class
 - D. All of the above
17. Regarding parameters passing in Java, which of the following statements is the best description?
- A. Only primitive and simple data types are passed by reference
 - B. All parameters are passed by value
 - C. Only object variables are passed by reference
 - D. All parameters are passed by reference
18. A group of objects is also known as what?
- A. Collection
 - B. Graph
 - C. Data set
 - D. Array
19. A hierarchical collection data structure is best represented by which of the following explanations?
- A. Ordered in a fashion where any element can have multiple successor elements and one predecessor element
 - B. Ordered in a fashion where so that any element can only have one successor element
 - C. Ordered in a fashion where any element can have multiple successor elements and also multiple predecessor elements
 - D. It stores the elements in a specific order and has repeated elements
20. A bag data structure is best represented by which of the following explanations?
- A. It can only access the last element in the data structure
 - B. It is ordered and cannot have duplicate elements
 - C. It is unordered and can have duplicated elements
 - D. It is unordered and cannot have duplicate elements

21. A queues data structure is best represented by which of the following explanations?
- A. It can only access the last element in the data structure
 - B. It is allowed access any element in the data structure
 - C. It can only access the first element in the data structure
 - D. It is unordered and cannot have duplicate elements
22. Choose the collection class below that allows duplicated elements in its data:
- A. TreeSet
 - B. RedundantList
 - C. HashSet
 - D. ArrayList
23. Arrays can store a tremendous amount of information, which of the following can they store?
- A. Arrays
 - B. Primitive data types
 - C. Objects
 - D. All of the above
24. Which of the following data structures starts with a root node that can be linked to one or two more nodes?
- A. Binary Search Tree
 - B. Binary Tree
 - C. Binary Hash Tree
 - D. Tree
25. When a method calls itself to solve a problem, is known as what type of process?
- A. Iteration
 - B. Compiling
 - C. Recursion
 - D. Hash Mapping
26. When a recursive method and an iterative method that perform the same task, what can be said about the executions speed?
- A. The iterative method is slower than the recursive method.
 - B. The iterative method is faster than the recursive method.
 - C. They execute at the same speed.
 - D. Sometimes iterative is faster depending upon the amount of cache memory
27. Comparing a recursive method and an iterative method that performs the same task, what can be said about the amount of memory needed for parameters and variables?
- A. You cannot compare the memory requirements of these two methods
 - B. A recursion method uses the same amount of memory as an iterative method
 - C. An iterative method uses more memory
 - D. An iterative method uses less memory

28. All things considered, which of the following scenarios will a merge sort perform best?
- A. 2 lists that are the same size and sorted
 - B. 2 lists that are the different sizes and sorted
 - C. 2 unsorted lists
 - D. Random lists
29. Which of the following best describes mutual recursion?
- A. A single method calls itself
 - B. An iterative method calls a recursive method that calls another iterative method
 - C. One method calls a second another method, and the second method calls the first method
 - D. All of the above
30. Consider a Binary Search algorithm: inferring the quantity of data processed doubled. What will be the effect upon the elapsed time of the function?
- A. The elapsed time will not be effected
 - B. There will be an insignificant increase in elapsed time
 - C. There will be a quadrupling of the elapsed time
 - D. The elapsed time will double
31. Consider an algorithm that counts from 1 to N: inferring the quantity of data processed doubled. What will be the effect upon the elapsed time of the function?
- A. The elapsed time will not be effected
 - B. There will be an insignificant increase in elapsed time
 - C. There will be a quadrupling of the elapsed time
 - D. The elapsed time will double
32. Consider an algorithm that computes the factorial of N (i.e. $N!$): inferring the quantity of data processed doubled. What will be the effect upon the elapsed time of the function?
- A. The elapsed time will not be effected
 - B. There will be an insignificant increase in elapsed time
 - C. There will be a quadrupling of the elapsed time
 - D. The elapsed time will double
33. Consider an algorithm that deletes an element from the end of an array: inferring the quantity of data processed doubled. What will be the effect upon the elapsed time of the function?
- A. The elapsed time will not be effected
 - B. There will be an insignificant increase in elapsed time
 - C. There will be a quadrupling of the elapsed time
 - D. The elapsed time will double

34. An algorithm has $O(n)$ complexity in terms of time. How long will it take to complete if $n = 3000$, if the execution of the algorithm takes 0.05 seconds if $n = 1000$?
- A. 0.10 seconds
 - B. 0.15 seconds
 - C. 0.30 seconds
 - D. 1.00 seconds
35. An algorithm has $O(1)$ complexity in terms of time. How long will it take to complete if $n = 3000$, if the execution of the algorithm takes 0.05 seconds if $n = 1000$?
- A. 0.05 seconds
 - B. 0.10 seconds
 - C. 0.15 seconds
 - D. 0.20 seconds
36. The software development cycle includes which of the following stages?
- A. Designing
 - B. Programming
 - C. Testing
 - D. All of the above #
37. Which definition below best explains a Data Structure Oriented Design?
- A. A technique of modular decomposition to transform data structures into a representation of software
 - B. A technique of modular decomposition to solve solutions through the creation of object and operations
 - C. A method of creating modular based upon the movement of information from input to file storage and report generation
 - D. A method of disseminating cross modular packet displacement for block chain node support
38. What do software engineers use for development environment for building applications, applets, and components using the Java programming language?
- A. JAR
 - B. JRE
 - C. JVM
 - D. JDK
39. Which of the following is included in the JDK
- A. JRE
 - B. Compiler
 - C. Debugger
 - D. All of the above

40. Which of the following is a characteristic of the Agile software development methodology?
- A. Based upon an iterative process
 - B. Based upon a systematic and sequential process where development moves unilateral direction
 - C. Progression system that uses virtual machines to stabilize deployment
 - D. Manages project scope creep for targeted latency
41. The style of programming and the strategy applied for a particular language is known as what?
- A. Programming Paradigm
 - B. Programming Production
 - C. Programming Development
 - D. Programming Strategy
42. Regarding ULM, which of the following represents a type of Behavior Diagram?
- A. Activity Diagram
 - B. Component Diagram
 - C. Sequence Diagram
 - D. Class Diagram
43. What shape in a flowchart indicates the movement from one component to another?
- A. Arrow
 - B. Rectangle
 - C. Diamond/ Rhombus
 - D. Parallelogram
44. Choose the SQL script below that will allow the program to choose all of the columns from “used_cars” list for people who bought a “Ford.”
- A. `SELECT ALL FROM used_cars WHERE item = 'Ford';`
 - B. `SELECT * FROM used_cars WHERE item = 'Ford'; #`
 - C. `SELECT “Ford” FROM used_cars WHERE *;`
 - D. `FROM used_cars SELECT item = “Ford” COLUMN *;`
45. Choose the SQL script below that will allow the program to choose distinct “item” from the “recentOrders” table:
- A. `SELECT DISTINCT item FROM items_ordered;`
 - B. `SELECT item FROM items_ordered;`
 - C. `SELECT DISTINCT items_ordered FROM item;`
 - D. `SELECT item FROM items_ordered DISTINCT item != item[i+1];`

46. What will be displayed by the method call *readyBPA* (434,232)?

```
public static int readyBPA(int x, int y)
{
    if (x / y >= 0)
    {
        System.out.print(x + " ");
        return y;
    }
    else
    {
        System.out.print(y + " ");
        return readyBPA(y, x % y);
    }
}
```

- A. 345
- B. 434
- C. Error message
- D. Nothing will print

47. What will be displayed by the method call *readyBPA* (5, 34)?

```
public static int readyBPA(int x, int y)
{
    if (x == 1)
    {
        print( y);
    } else {
        print(" "+ x);
        return readyBPA (x-1);
    }
}
```

- A. 5 4 3 2 1
- B. 5 4 3 2 1
- C. Error message
- D. Nothing will display

48. What value is stored by the statement: `double ans = 121.0/5?`

- A. 24.2
- B. 24.0
- C. 24
- D. Error message

49. What value is stored by the statement: `double d = (double)((int)121.0/(int)5.0)?`

- A. 24.2
- B. 24.0
- C. 24
- D. Error message

50. What value is stored by the statement: `int d = ((int)-121.1 % 3 % (int)17.0)`

- A. -1
- B. -1.0
- C. 1
- D. 1.0